Domain Testing Report

Document control

Field	Value
Student/group	`10`
Date updated	`2026-10-18`
Scope	FR-05, FR-11, FR-19, and selected FR-20 subsection
Method	Equivalence Partitioning and Domain Testing

Rules and evidence

- Classes below are based on the checked feature inventory and source files.
- A class is marked `Unresolved` where the source/specification does not establish a constraint or oracle.
- Valid domain tests may multi-cover valid classes. Invalid tests vary one invalid class while keeping other variables nominal.
- `Expected output` means the observable result to verify; implementation behavior that conflicts with the requirement is not silently treated as correct.

Evidence ID	Source/route/file	Verified observation
E-001	`README.md (of the project)`, FR-05	Product listing/search supports loading and empty-result states.
E-002	`frontend-web/src/pages/Home.jsx`	Empty search requests all products; non-empty search submits `search`; result count appears for non-empty results.
E-003	`frontend-mobile/App.js`	Mobile product flow has loading, error, product-list, search, and empty-array branches.
E-004	`backend/server.js`, `GET /api/orders/my-orders`	Authenticated orders are filtered by `req.user.id` and sorted by descending ID.
E-005	`frontend-mobile/App.js`, `renderProfile`	Mobile history shows order ID, localized date, formatted total, mapped status, and an empty-order message.
E-006	`README.md (of the project)`, FR-19	Admin may view users without passwords and delete users except the current administrator.
E-007	`backend/server.js`, `/api/admin/users`	User list query omits password; inspected admin endpoints use authentication middleware but show no explicit role/current-admin check.
E-008	`frontend-admin/src/App.jsx`	Admin UI has user list and delete action; login UI checks the returned role before storing the token.
E-009	`frontend-mobile/App.js`, mobile product/history flow	FR-20 selected scope reuses mobile browsing and authenticated order history; cancellation is excluded.

1. FR-05 - Product listing and search

Requirement and scope

- In scope: displaying products, submitting a product-name search, and observable loading, error, and empty-result states.
- Out of scope: product detail, cart, checkout, and product administration.
- Search maximum length, accepted character set, and dedicated no-results message are not defined by the checked evidence.

Input/output variables

ID	Type	Variable	Evidence-backed domain
FR05-V01	Input	Search value	Empty value; non-empty text. Maximum length/character restrictions unresolved.
FR05-V02	State	Product/request state	Loading; successful response with products; successful response with no products; error response.
FR05-O01	Output	Product result	Product cards/list; empty result rendering; error rendering.

Complete input/output equivalence classes

EC ID	Variable	Valid/invalid	Condition/domain	Representative	Expected observable output	Evidence	Review
FR05-EC-01	Search value	Valid	Empty search	""	All available products are requested/displayed.	E-001,E-002	Pending
FR05-EC-02	Search value	Valid	Non-empty text matching one or more names	A verified product-name fragment	Matching products are returned/displayed.	E-001,E-002	Pending
FR05-EC-03	Search value	Valid	Non-empty text matching no product	Verified absent name fragment	Empty result is observable; exact message remains unresolved.	E-001,E-002	Pending
FR05-EC-04	Product/request state	Valid state	Loading request	Controlled delayed request	Loading indicator/state is observable where implemented.	E-001,E-003	Pending
FR05-EC-05	Product/request state	Valid state	Request/database error	Controlled error response	Error state is observable.	E-001,E-002,E-003	Pending
FR05-EC-06	Search value	Invalid/unresolved	Unsupported length/characters	`[Value after constraint is confirmed]`	Cannot define an oracle until a UI/API constraint is verified.	E-002	Pending

Minimum domain test cases

ID	ECs covered	Representative data	Single-fault check	Expected output	Traceability
DT-FR05-01	EC-01	Empty search, nominal product data	Valid nominal	Product list is displayed.	FR-05/EC-01
DT-FR05-02	EC-02	Matching product-name fragment	Valid nominal	Matching product cards are displayed.	FR-05/EC-02
DT-FR05-03	EC-03	Absent product-name fragment	Valid nominal	Empty result is displayed without matching cards.	FR-05/EC-03
DT-FR05-04	EC-04	Delayed product response	Valid nominal	Loading state is observable during request.	FR-05/EC-04
DT-FR05-05	EC-05	Controlled request error	One error state only	Error state is displayed.	FR-05/EC-05

2. FR-11 - User order history

Requirement and scope

- In scope: authenticated users viewing only their own orders, including ID, date, total, and translated current status.
- Out of scope: checkout, admin order management, and mobile cancellation.
- The order-detail endpoint exists without authentication middleware in the inspected source; its security oracle requires explicit runtime/spec review.

Input/output variables

ID	Type	Variable	Evidence-backed domain
FR11-V01	State	Authentication	Valid authenticated token; missing/invalid token.
FR11-V02	State	Order count for current user	Zero; one; multiple orders.
FR11-V03	Data	Order status	`pending`, `confirmed`, `shipping`, `delivered`, `canceled`; unknown status fallback is implementation behavior.
FR11-O01	Output	History contents	Own orders only, descending ID; ID/date/total/status fields; empty state when no orders.

Complete input/output equivalence classes

EC ID	Variable	Valid/invalid	Condition/domain	Representative	Expected observable output	Evidence	Review

EC ID	Variable	Valid/invalid	Condition/domain	Representative	Expected observable output	Evidence	Review
FR11-EC-01	Authentication	Valid	Authenticated current user	Valid token for User A	User A's history can be loaded.	E-004,E-005	Approved
FR11-EC-02	Authentication	Invalid	Missing or invalid token	No token / invalid token	Access is rejected rather than returning a history.	E-004	Approved
FR11-EC-03	Order count	Valid	No orders for current user	User A with zero orders	Mobile empty-order message is displayed.	E-005	Approved
FR11-EC-04	Order count	Valid	Exactly one own order	User A with one order	One order with required fields is displayed.	E-004,E-005	Approved
FR11-EC-05	Order count	Valid	Multiple own orders	User A with two or more orders	All own orders are displayed in descending ID order.	E-004,E-005	Approved
FR11-EC-06	Order status	Valid	Supported status value	One order per supported status	Status is translated to the observed Vietnamese label.	E-005	Approved
FR11-EC-07	Ownership	Invalid/security	Attempt to obtain another user's order through own-history flow	User A token, User B order	User B's order must not appear in User A history.	E-004	Approved

Minimum domain test cases

ID	ECs covered	Representative data	Single-fault check	Expected output	Traceability
DT-FR11-01	EC-01,EC-04,EC-06	User A, one own order, `pending`	Valid multi-cover	One own order with ID/date/total/status is shown.	FR-11/EC-01,04,06
DT-FR11-02	EC-01,EC-05,EC-06	User A, multiple own orders with supported statuses	Valid multi-cover	All own orders are shown in descending ID order with translated statuses.	FR-11/EC-01,05,06
DT-FR11-03	EC-01,EC-03	User A, no orders	Valid multi-cover	Empty-order state is shown.	FR-11/EC-01,03
DT-FR11-04	EC-02	Missing/invalid token	One invalid class only	Request is rejected; no order history is returned.	FR-11/EC-02
DT-FR11-05	EC-07	User A token, User B order data	One ownership violation only	User B order is not exposed through User A history.	FR-11/EC-07

3. FR-19 - Admin user management

Requirement and scope

- In scope: authorized admin viewing users without passwords and deleting a user other than the currently logged-in administrator.
- Out of scope: admin authentication implementation, role administration, bulk selection, and unrelated admin modules.
- The inspected backend does not prove role authorization or current-admin protection; these remain verification targets, not assumed results.

Input/output variables

ID	Type	Variable	Evidence-backed domain
FR19-V01	State	Authentication/role	Admin; ordinary user; missing/invalid token.
FR19-V02	State	Customer-user-list cardinality, excluding the logged-in admin baseline	Zero; one; multiple customer users.

ID	Type	Variable	Evidence-backed domain
FR19-V03	Input	Target user identity	Other user; current admin; nonexistent/malformed identifier.
FR19-O01	Output	User list/delete result	User data excluding password; successful deletion or protected/error result.

Complete input/output equivalence classes

EC ID	Variable	Valid/invalid	Condition/domain	Representative	Expected observable output	Evidence	Review
FR19-EC-01	Auth/role	Valid	Authenticated admin	Admin token	User management is accessible.	E-006,E-008	Approved
FR19-EC-02	Auth/role	Invalid	Ordinary-user token	User token	Admin operation is rejected.	E-006,E-007	Approved
FR19-EC-03	Auth/role	Invalid	Missing/invalid token	No token / invalid token	Operation is rejected.	E-007	Approved
FR19-EC-04	Customer-user count	Valid	Zero customer users, with admin baseline retained	Dataset with admin only	Admin baseline is listed; no customer user is listed.	E-006,E-008	Approved
FR19-EC-05	Customer-user count	Valid	One or more customer users	Dataset with one/multiple customer users	Customer users are listed without password values.	E-006,E-007,E-008	Approved
FR19-EC-06	Target identity	Valid	Other user's existing ID	Existing non-admin user ID	Target user is deleted and list refreshes.	E-006,E-008	Approved
FR19-EC-07	Target identity	Invalid/protected	Current admin ID	Logged-in admin ID	Current administrator is not deleted.	E-006	Approved
FR19-EC-08	Target identity	Invalid	Nonexistent or malformed ID	Verified absent/malformed ID	Error/no-op behavior must be verified; source does not establish it.	E-007	Approved

Minimum domain test cases

ID	ECs covered	Representative data	Single-fault check	Expected output	Traceability
DT-FR19-01	EC-01,EC-05	Admin token, one/multiple users	Valid multi-cover	Users are listed and no password field is exposed.	FR-19/EC-01,05
DT-FR19-02	EC-01,EC-04	Admin token, zero customer users with admin baseline	Valid multi-cover	Admin baseline is shown and no customer user is listed.	FR-19/EC-01,04
DT-FR19-03	EC-01,EC-06	Admin token, other user's existing ID	Valid action	Other user is deleted and result is observable.	FR-19/EC-01,06
DT-FR19-04	EC-02	Ordinary-user token	One invalid class only	Admin operation is rejected.	FR-19/EC-02
DT-FR19-05	EC-03	Missing/invalid token	One invalid class only	Operation is rejected.	FR-19/EC-03
DT-FR19-06	EC-01,EC-07	Admin token, current admin ID	Protected target only	Current admin remains.	FR-19/EC-01,07
DT-FR19-07	EC-08	Admin token, invalid/nonexistent ID	One invalid class only	Verified error/no-op behavior is recorded.	FR-19/EC-08

Gate A review

Approved by the student after evidence reconciliation. The student confirmed the FR-19 equivalence classes, representatives, and minimum domain tests, including the distinction between zero customer users and the retained admin baseline.

4. FR-20 - Mobile product browsing and order history

Requirement and scope

- In scope: mobile product listing/search and authenticated mobile order history.
- Out of scope: mobile authentication, profile editing, cart, checkout, and order cancellation.
- Mobile browsing reuses the product-state classes from FR-05; mobile history reuses the ownership/count/status classes from FR-11.

Input/output variables

ID	Type	Variable	Evidence-backed domain
FR20-V01	Input/state	Mobile search/product request	Empty/non-empty search; loading; products; no products; error.
FR20-V02	State	Mobile authenticated history	Authenticated current user; no/one/multiple own orders.
FR20-O01	Output	Mobile browsing/history	Product cards/search result state; order ID/date/total/translated status; empty history state.

Complete equivalence classes

EC ID	Variable	Valid/invalid	Condition/domain	Representative	Expected observable output	Evidence	Review
FR20-EC-01	Product search	Valid	Empty search	""	Mobile product list is loaded.	E-003,E-009	Pending
FR20-EC-02	Product search	Valid	Matching search	Verified product fragment	Matching mobile products are shown.	E-003,E-009	Pending
FR20-EC-03	Product result	Valid	No matching product	Absent fragment	Empty product result is observable.	E-003	Pending
FR20-EC-04	Product request	Valid state	Loading/error	Delayed/error request	Loading or error state is observable.	E-003	Pending
FR20-EC-05	History auth	Valid	Authenticated user	Valid token	Current user's history is loaded.	E-004,E-005,E-009	Pending
FR20-EC-06	History count	Valid	Zero/one/multiple own orders	Dataset cardinality	Empty state or all own orders are shown.	E-005,E-009	Pending
FR20-EC-07	History ownership	Invalid/security	Other user's order	User A token, User B order	Other user's order is not shown.	E-004,E-009	Pending

Minimum domain test cases

ID	ECs covered	Representative data	Single-fault check	Expected output	Traceability
DT-FR20-MOB-01	EC-01	Empty search, nominal product data	Valid nominal	Mobile product list is displayed.	FR-20/EC-01
DT-FR20-MOB-02	EC-02	Matching product fragment	Valid nominal	Matching products are displayed.	FR-20/EC-02
DT-FR20-MOB-03	EC-03	Absent product fragment	Valid nominal	Empty product result is observable.	FR-20/EC-03
DT-FR20-MOB-04	EC-04	Delayed/error product request	One request state	Loading/error state is observable.	FR-20/EC-04
DT-FR20-MOB-05	EC-05,EC-06	Authenticated user with one/multiple orders	Valid multi-cover	Own order history fields and status labels are shown.	FR-20/EC-05,06
DT-FR20-MOB-06	EC-05,EC-06	Authenticated user with no orders	Valid multi-cover	Empty history message is shown.	FR-20/EC-05,06
DT-FR20-MOB-07	EC-07	User A token, User B order data	One ownership violation only	User B order is not shown.	FR-20/EC-07

Gate A review

Gate item	Status	Student review/correction
All classes have evidence or are marked unresolved	Approved	Student approved on 2026-10-18; no corrections
Representatives are executable	Approved	Student approved on 2026-10-18; no corrections
Invalid cases follow single-fault principle	Approved	Student approved on 2026-10-18; no corrections
Gate A approved before BVA	Approved	Student approved on 2026-10-18; no corrections

FR-11 Gate A was additionally reviewed by the student on 2026-10-19. No corrections or missing domain edges were reported.